



Breakthrough
Energy

Industry Perspectives on Uncertainty and Integrated Planning

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PowerUp 2026 — Planning Under Uncertainty

Four Stops, One Fundamental Pursuit





Uncertainty Is Stacked, Not Single-Axis

Compounding, not additive — and exactly the field this room works in

LOAD & DEMAND

Electrification, data centers, reshoring — demand curves that used to move a percent a year now move in double digits.

TECHNOLOGY COST

Solar and battery costs keep falling. The policy support underneath some of that curve doesn't always fall with it.

POLICY & POLITICS

Tax credits, permitting rules, and market design can reverse between one administration and the next.

WEATHER & SUPPLY CHAIN

Interconnection queues, mineral supply, and multi-week low-wind, low-sun events stack on top of everything else.



An impossible problem?



A once in a lifetime decision...

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What about operations?



The “Moneyball” Problem

What the model optimizes vs. what the operator’s console shows

MWh $\neq$ capacity value. Being available at the hour that matters isn’t in most objective functions.

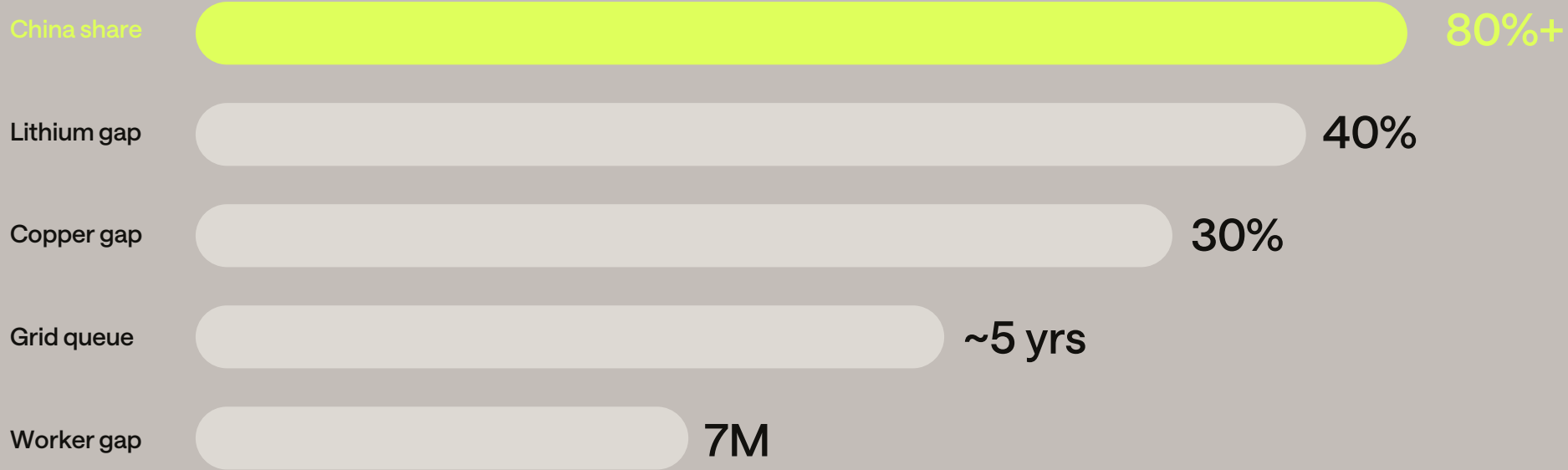
Falling tech cost $\neq$ falling system cost. Inertia, reactive power, and ramp capability get more expensive as penetration rises.

Interconnection $\neq$ paperwork. 2,600 GW is queued in the U.S. alone; the average wait is nearly five years.



A plan is only
as good as its
potential execution.

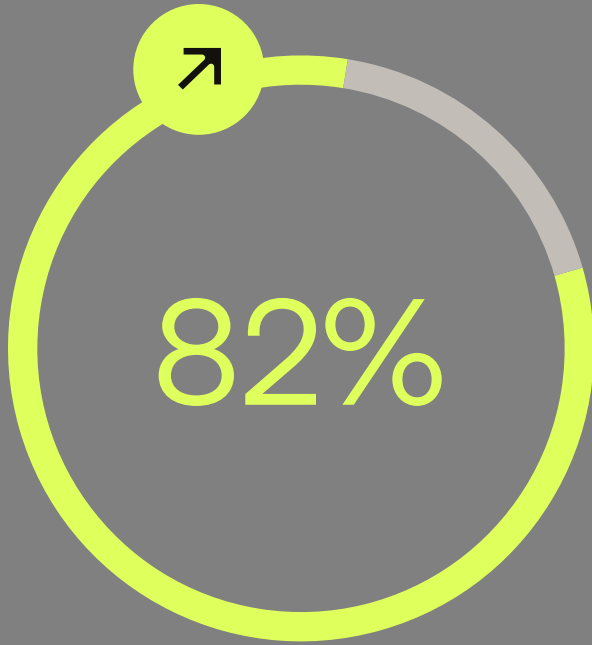




Supply Chain & Workforce Reality

Five numbers that decide how fast any pathway can actually be built

None of these gaps need a technology breakthrough. All of them need investment made years before the need shows up.



The strongest predictor of acceptance isn't economics. It's whether a community felt consulted early, and genuinely.

THE DANISH PARADOX

of Danes support wind energy. A local 20% cancelled the vast majority of projects in two years. Minority veto power is the structural problem.



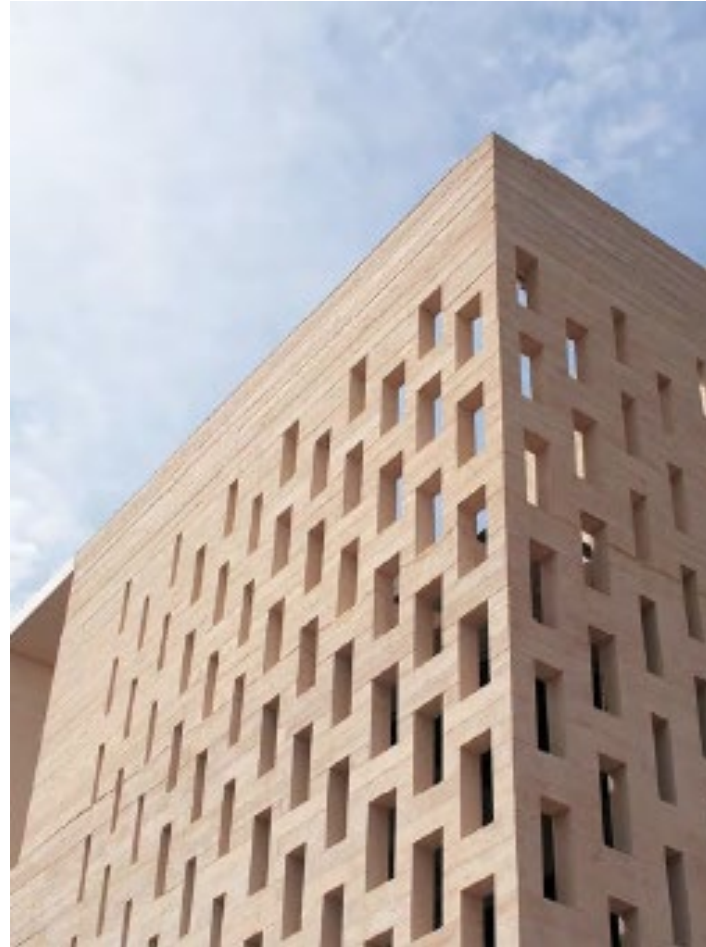
The Conversation at the Kitchen Table

Where a plan actually gets accepted, or doesn't



What I'd Ask of This Room

Your models are the connective tissue between political will and physical reality



KEEP THIS IN THE MATH

- + Model the execution risk, not just the resource cost — supply chains, workforce, and permitting timelines are inputs too.
- + Build tools fast and legible enough to earn trust before the decision window closes.
- + Remember someone will have to live with what the model recommends. Build for that person, not just the objective function.



A black and white photograph of a man in a lab coat and glasses, looking intently at a complex mechanical device. The device has many wires and components, suggesting a scientific or engineering context. The man is on the right side of the frame, and the device is on the left.

Quote attributed to Winston Churchill

**“Plans are of little
importance, but planning
is essential.”**



Thank you.



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